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Structuring a Program in Elite Professional Soccer

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ABSTRACT

THIS ARTICLE DISCUSSES THE PROCESSES THAT ARE UNDERTAKEN BY THE CONDITIONING COACHES OF ELITE SOCCER CLUBS WHEN PLANNING THE TRAINING PROGRAM FOR PROFESSIONAL PLAYERS. TO SUCCESSFULLY MANAGE THE TRAINING REQUIREMENTS OF A SQUAD OF PLAYERS THROUGHOUT A SEASON, IT IS IMPORTANT TO STRUCTURE TRAINING APPROPRIATELY USING STRATEGIES TO IDENTIFY THE INDIVIDUAL NEEDS OF PLAYERS, ENSURING THAT THOSE REQUIREMENTS ARE MET WITHIN THE CONSTRAINTS OF THE ANNUAL CALENDAR. THE OVERALL AIM IS TO MAXIMIZE SQUAD AVAILABILITY WHILE OPTIMIZING PLAYER READINESS FOR MATCHES, ENSURING THAT EACH PLAYER HAS THE PHYSICAL CAPABILITY TO IMPLEMENT THE REQUIRED TACTICS SUCCESSFULLY.

INTRODUCTION

Periodization is the strategic manipulation of training stress to produce a desired outcome, leading to competitive success. A number of strategies have been proposed to maximize training adaptation in athletes, such as linear, conjugate, concurrent, block, and concentrated models (14); however, there are a number of factors that affect an individual's

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training response, such as genetic inheritance, training history, biological functioning, and stress, both psychosocial and environmental; it is understood that there is no "one size fits all" approach because individual athletes respond differently to one another. Furthermore, identical training sessions performed by the same individual may elicit different responses due to the transient nature of biological functioning and the environmental stressors that athletes face.

With regard to professional soccer, weekly models tend to vary based on culture and previous experiences of the management team. Typically, plans are based around a program, whereby there are 6–7 days between competitive matches; however their applicability to successful teams is limited, and they are not appropriate for dealing with the real planning issues that elite teams must overcome. The extensive nature of the soccer annual calendar presents a number of challenges because players of Europe's elite teams participate in up to 220 training sessions and 60 competitive matches per season. Recently, the practice of "strategic periodization" has become popular in some sports (23), whereby a match difficulty index is created for each match during the season; thus, training volume and intensity can be manipulated weekly by the coaching staff to allow for intentional peaking for matches of perceived greatest priority or importance throughout a season. However, an analysis of the number of matches lost for the English Premier League winners throughout the previous 10 seasons showed that teams lost on average only 4.6 times per season; thus,

not going into all matches with optimal freshness and strategically placing greater importance on certain matches may be a risky practice for coaches.

The challenge at the elite level is clear and was highlighted previously by Iaia and Hawkins (12); they suggested that the primary purpose of fitness training is not pushing the boundaries of human performance and sustaining the highest work rates, but it is more oriented toward keeping the players injury free and making them capable of playing competitive games every 3–4 days for most of the season throughout their whole career. The importance of this was demonstrated by Carling et al. (6) who reported a correlation between player squad availability and match success in an elite French club for more than 5 seasons. By remaining injury free, the coach was able to select similar starting lineups more frequently than in other seasons, presumably playing with his best team more often, leading to greater tactical and technical understanding from the players with successful performance outcomes. To achieve this, consideration must be given to a multitude of factors that impact the strategies that underpin the decisions that are made on a daily basis over the course of a season.

Three key strategies form the basis of our planning model that is transient, depending on the philosophy of the coach and the individual needs of each player. The coach's philosophy is of

KEY WORDS:

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paramount importance because the formation employed, style and tactics of play, in attack, defense, and transition will affect the physical activity profiles of each position and individual player during a match. Understanding these profiles is fundamental to develop and optimize the physical preparation programs of elite players. The discussion that follows involves components that impact 1 or more of the 3 strategic pillars identified in Figure 1. Ultimately, by maximizing player availability, ensuring that players have the physical capabilities to carry out the coach's instructions and that they begin each match in an optimal state of readiness, everything has been done from a physical perspective to enhance the probability of success.

We will begin by discussing planning at the annual level before focusing on the weekly plan and ultimately drilling into the daily training themes and examples of specific exercises. The importance of off-field conditioning is also discussed, and how this is integrated into the planning process and finally how issues related to squad management can be addressed.

ANNUAL CALENDAR

When planning conditioning programs for soccer players, the annual calendar is divided into 3 phases:

preseason or preparation, in-season or competition, and off-season or transition. Each of these phases have specific goals and associated challenges with regard to structuring players' training. The seasonal structure in elite European soccer is markedly different from some sports, such as American (National Football League) or Australian Football (Australian Football League) that are characterized by 5- to 6-month seasons, with a long transition/off-season phase in the National Football League and a long preseason phase in the Australian Football League to allow time to develop physical qualities before the commencement of the competitive season. Elite players of Europe's soccer teams typically experience a 5- to 6-week preseason, followed by a 41- to 42-week in-season with extensive national and international travel commitments for teams participating in both domestic and European Competitions. Frequently, teams that are new to entering European competitions have demonstrated an inability to manage a debut European season together with maintaining domestic form. An additional potential confounder is the off-season "summer period" and whether players are involved in an international tournament. This affects the decisions made with regard to the recovery/restoration period players require before returning to preseason training or in some cases after the start of the new season. In these cases, international players

frequently receive only 21–28 days off before returning for a new season.

A further hurdle with regard to planning preseason training is the tours that elite teams frequently embark on before the competition phase. Europe's elite teams now routinely travel to America, Asia, Africa, and Australasia to play a number of matches, with the first taking place approximately 10 days after the commencement of preseason training. In addition, there is a growing tendency for high-profile clubs to participate in "showcase" friendlies around the world after the end of the competitive season, further increasing the demands placed upon players. Both preseason and off-season training are beyond the scope of this article, so the remainder will discuss the considerations that take place with regard to planning training during the in-season period.

WEEKLY STRUCTURE

Designing the ideal training week involves the consideration of numerous sources of information (Figure 2). Generally, the aim is to overload game components in such a way that advances in fitness are made or simply maintained, without compromising readiness for the next game. Flexibility is required in any plan: the combination of objective data, experience and intuition driving processes, and a successful outcome blending both art and science.

The task is relatively simple when 5–7 days exist between competitive games; however, the number of these "free weeks," weeks in which there is no midweek game, are few and far between during successful seasons; there may be in the region of 4–6 weeks during the season that there is an opportunity to apply such a model. Figure 3 depicts a potential weekly plan through which the majority of player conditioning requirements can be appropriately targeted, ensuring adequate stimulation to maximize competition readiness. However, the addition of a midweek game typically

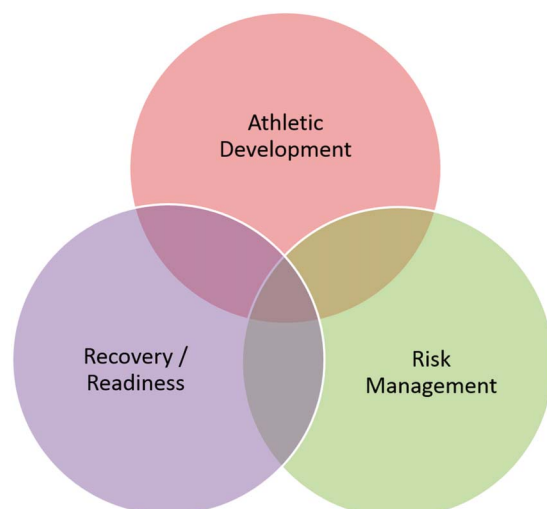


Figure 1. Three key strategic planning pillars.



Figure 2. Considerations when planning player physical training needs.

results in only 1 day to focus on conditioning for players who were not involved in the previous game. In this situation, the requirements of individual players must be considered, as well as the conditioning focus for that particular stage of the season.

DAILY TRAINING THEME

It is clearly evident that time is limited and that the available time that players can be exposed to training must be

effectively utilized to ensure that the desired goals are met; the luxury of a 2-day recovery period, 2-day acquisition period, and a 2-day tapering period leading into the next game is rarely a reality.

There are various training categories that can be combined and incorporated into a daily training theme whereby the majority of conditioning components can be targeted alongside tactical preparation requirements (Figure 4). Developing daily plans

and integrating them into a weekly structure enables players to be trained at an appropriate level, avoiding training monotony, while also ensuring recovery and optimizing player readiness for matches. Although there is a requirement for flexibility, the training categories identified can be conveniently grouped to produce a daily theme.

The 2 key daily themes utilized can be classified as “extensive soccer” and “intensive soccer” and incorporate the following:

- **Extensive soccer** aims to reflect typical match demands, utilizing large areas enabling high speeds to be attained. Speed components can also be added to the beginning of the session with specific speed work or soccer-specific drills, such as speed finishing or wave attacks being utilized.
- **Intensive soccer** overloads the musculoskeletal system and the various energy systems, incorporating more acceleration, deceleration, and change of direction components of soccer-specific speed. This area incorporates small-sided games, which are modified games played in reduced playing areas, using fewer players and adapted rules (10). By manipulating pitch dimensions, number of players, rules, coach encouragement, goalkeepers, and work to rest ratios, the intensity of training can be appropriately manipulated. Depending on the stage of the season and the requirements of individual players, these sessions can be supplemented with various conditioning runs if required.

A pertinent discussion here involves the incorporation of off-field strength and conditioning practices into the weekly planning template, presented in Figure 3. Depending on the coach philosophy and the strength and conditioning culture at the club, players are prescribed off-field conditioning to improve their strength, power, or resilience/robustness as appropriate. A detailed description of off-field conditioning will follow, but it is important that off-field strength and

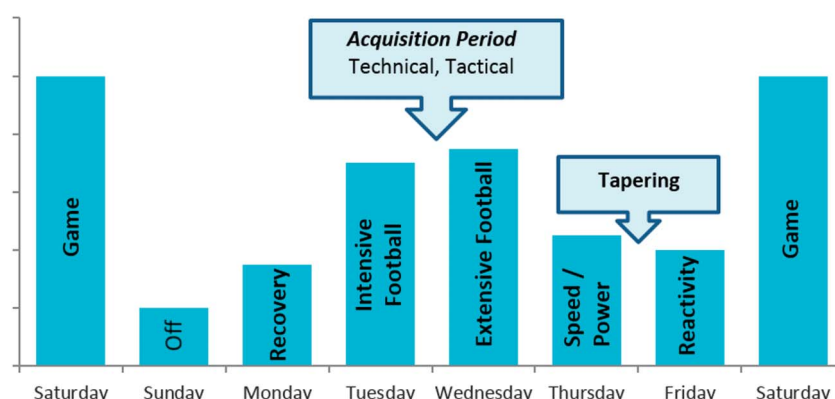


Figure 3. Potential weekly planning template.

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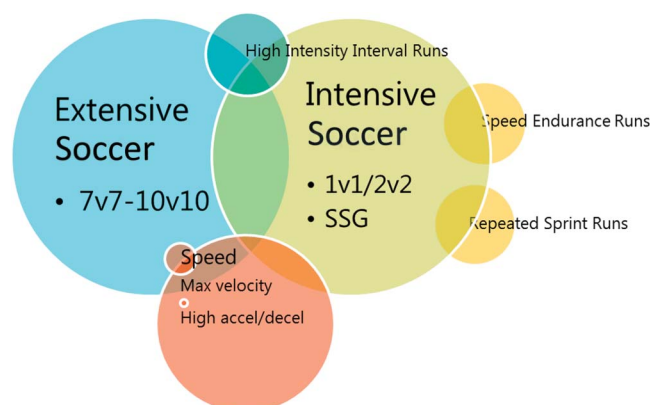


Figure 4. Training categories targeting specific conditioning components.

conditioning programs are carefully planned, based on a thorough needs analysis of each player and delivered in accordance with the athlete and coach's goals, together with the demanding fixture schedule. Good communication and liaison with the technical coaching staff is vital to combine off-field strength and conditioning work with on-field physical, technical, and tactical training, to minimize the interference effect of concurrent training (16) and to schedule training such that the player's risk of injury is not increased in subsequent sessions.

For example, in the on-field training scenario presented in Figure 3, if a player/squad performed a leg strength or injury prevention program consisting of exercises with high eccentric demand, such as the Nordic hamstring exercise, there may be an increased risk of injury when performing extensive soccer drills incorporating high-speed running elements during the following day's training (19). By rearranging the weekly planning template, extensive soccer conditioning would be conducted before the strength session on Tuesday with the intensive soccer session performed on

Wednesday, in which it is known that maximal running velocities are reduced on smaller-sized pitches. It is this constant communication and management of on-field and off-field training stimuli that is required to successfully integrate different physical conditioning components.

IDENTIFYING DRILL REQUIREMENTS

The requirement at the elite level from a physical perspective is to prioritize the individual requirements and implement the necessary drills to impose the desired demands to ensure that the player status is optimal. Field-based training can be utilized to induce the appropriate stressors to ensure the energy systems of players adapt and are best equipped to withstand the competitive demands of the game, presented in Table 1. Through an in-depth understanding of the training parameters taxed with certain drills, a better appreciation of the physiological objectives to target to stimulate the physical components of match performance is possible, ensuring that the training load imposed on individuals is appropriate for the intended physiological and performance adaptations.

Position	Total distance covered (m)	High-intensity distance (>19.8 km/h)	Sprint distance (>26.2 km/h)	No. of sprints	Six season increase in high-intensity running (%)	Other movement patterns (specific to that position)
Center back	9,896	612	153	36	33	More back, sideward movement, and jumping
Full back	10,730	1,115	288	60	35	More longer sprints
Center midfielder	11,495	953	217	55	30	More forward movements and 0–90° turns
Wide midfielder	11,612	1,214	331	68	27	More diagonal and curved movements
Attacker	10,320	1,026	312	55	24	Diagonal and curved runs, more 270–360° turns

Note that this contains average data, and there is high match-to-match variability for all positions due to factors, such as team tactics, ball possession, match importance, and status of match (i.e., winning or losing).

Coaching science plays an integral role in this process, taking advantage of the technological developments that have occurred to add a high level of objectivity to what was previously predominantly subjective. By implementing suitable monitoring strategies, the impact of sessions on individual player readiness can be predicted before the event. Also, the adoption of ongoing monitoring strategies help in reducing the injury risk by optimizing exposure to specific external load parameters; the challenge is to establish what the minimal effective dose is with regard to output in training to achieve the desired overload and protective effect, establishing whether they can be achieved via the planned drills that have a tactical and/or technical focus or whether there is a need for additional individual work.

There continues to be a debate surrounding this last point and how best to achieve the conditioning objectives of players. It is known that fitness performance in team sports does not rely predominantly on the physical components but is influenced by the technical, tactical, and psychological qualities (3). The development of fitness therefore, when possible, should be integrated into a holistic approach to training where the conditioning elements are developed simultaneously, in combination with the technical, tactical, and psychological components. This is the ideal, and it is possible to

achieve the desired aims when maintenance is required; however, when specific outcomes are desired and the aim is to trigger the specific physiological systems understood to be limiting factors to performance, holistic training whereby many mechanisms are taxed to lesser degrees can be found wanting, with successful outcomes attained by focusing on individuals and their specific positional demands. In most environments, a blend of both approaches is used, sometimes being driven by philosophical principles rather than player physiology and individual needs.

PLAYER REQUIREMENTS AND GAME PROXIMITY

The physical components that players possess cover a range of physiological systems, predominantly neurological, musculoskeletal, and metabolic, to varying degrees. There are many considerations to be undertaken when designing drills, and it is not possible to concentrate on every quality all of the time. **Priorities will change with different players, and positions and team tactical requirements are also likely to become a factor.**

Depending on the individual player, the requirements to develop a specific attribute will differ. Longitudinal monitoring of player activity utilizing the various technological systems available informs practitioners of what players have done, thereby assisting in the

determination of what training is required. Multiple decisions of this nature are constantly undertaken to identify what specific players need in the next session.

Figure 5 shows the potential thought process practitioners go through when planning specific drills and sessions. First, the proximity to the previous and the next game will dictate the template for the session, and the expected physical demands of the technical and tactical work planned need to be understood. Following this, decisions can be made for individuals. Those players who are not involved in the next game are usually returning from injury and tend to require greater emphasis on developing the various energy systems and explosiveness, whereas those involved in games generally fit into fewer categories as shown; there is a potential to exert a greater load on players in the early part of the training week, whereas when the next competitive game is within 48 hours, the emphasis is predominantly on quality rather than inducing an overload.

PLANNING FOR G-2/G-1

In-season, due to the scheduling and the number of games players are preparing for, it is often difficult to significantly develop all physical attributes. Such qualities however can be honed via minimal exposure, ensuring that players are suitably prepared for competition. **From a training perspective, it has been highlighted that any form of overload is difficult; however, short, explosive, high-intensity drills with small groups and/or individuals can provide a suitable stimulus, both mentally and physically.** Figure 6 provides an example of a repeated sprint drill designed for a center midfield player, and Figure 7 shows a speed/agility exercise performed with a striker. The intensity of the session is presented below, and the work to rest ratio is the same; however, the volume (number of sets and reps) will be dictated by the proximity to the previous and subsequent games, together with other variables presented in Figure 2.

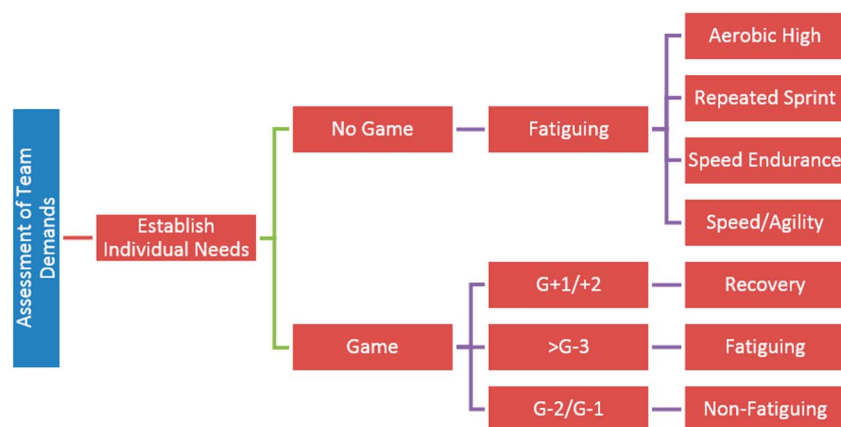


Figure 5. Identifying individual physical training focus.

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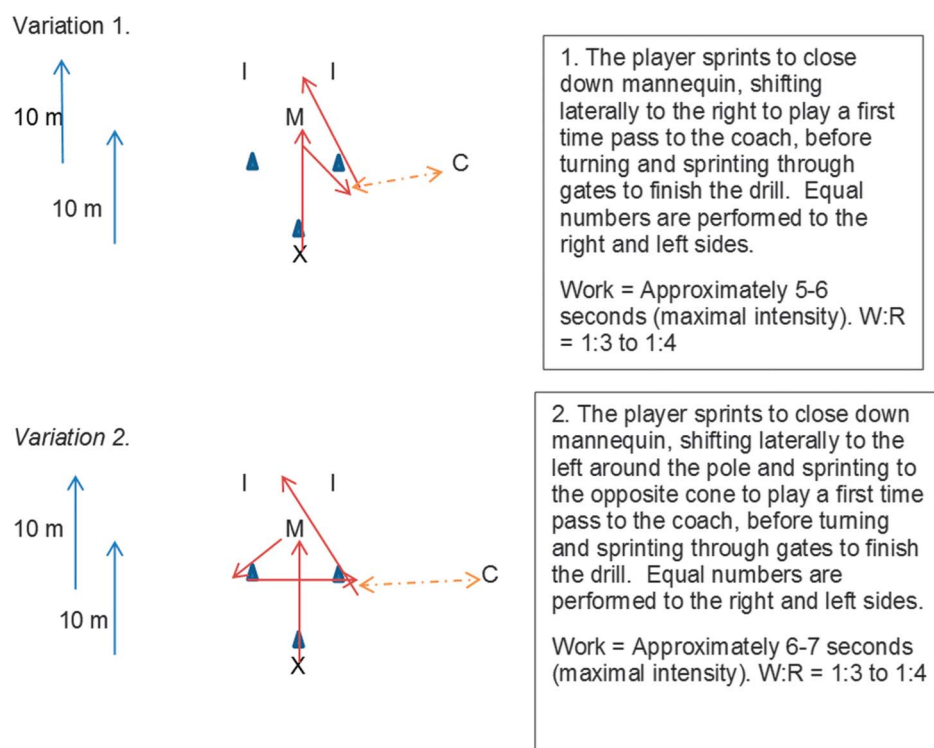


Figure 6. Repeated sprint drill for a center midfielder.

PLANNING FOR $\geq G-3$

When players are more than 72 hours before a game, it is usually possible to target any physical component. Depending on the content of the

technical and tactical work undertaken, knowledge of individual player profiles and individual needs analysis will direct the practitioner to the appropriate quality to overload. Explosivity is

fundamental at the top level and is associated with many of the attributes that elite players possess. It is therefore logical to target such qualities via high-speed/explosive-orientated drills rather

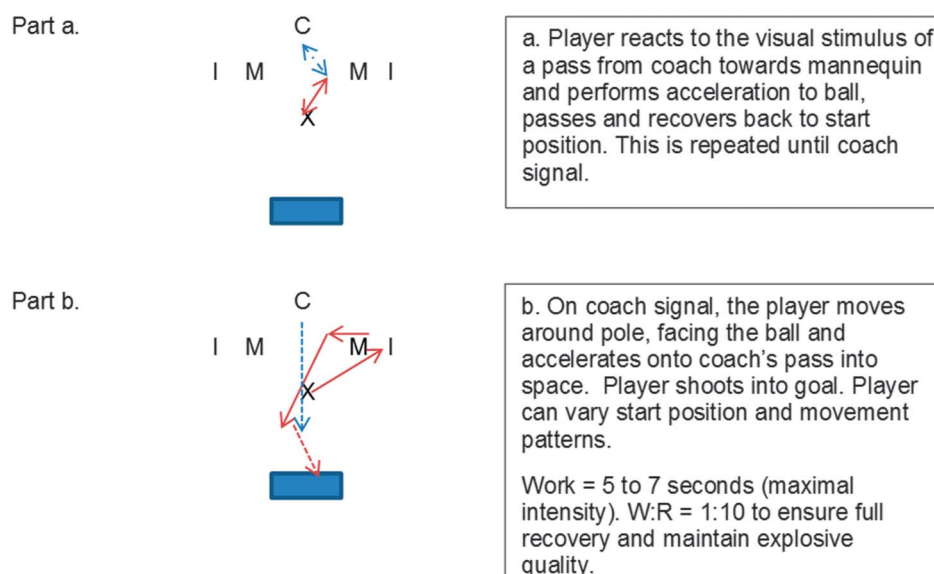


Figure 7. Reactive speed/agility drill for striker.

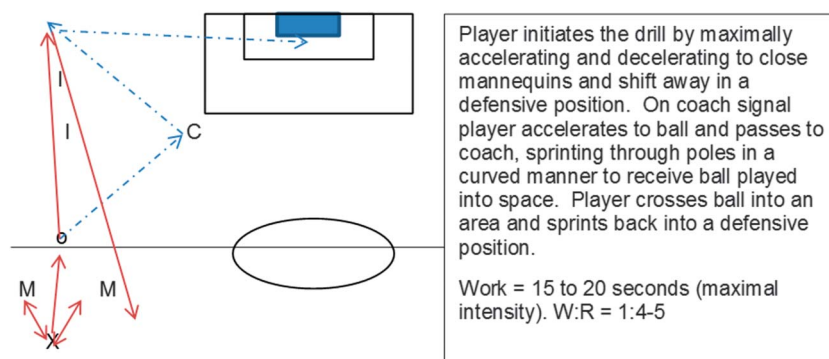


Figure 8. Speed-endurance drill for a full back.

than traditional steady-state running, even where energy system development is required; desirable adaptations can be attained via repetition of high-speed runs as described by Iaia and Bangsbo (11). Additionally, this type of training ensures that speeds that are commonly attained during elite match-play are replicated, as well as incorporating high accelerations and decelerations both of which have a role in injury prevention (7). Figure 8 demonstrates an example of a speed-endurance drill designed for a full back, and Figure 9 provides a high-intensity

aerobic endurance drill, typically performed with a center back. As previous, the volume of the session will be dictated by individual factors outlined in Figures 2 and 5.

OFF-FIELD CONDITIONING

Match analysis of modern elite soccer players has demonstrated that the current game demands a greater incidence of explosive events than previously. Increased muscle strength is associated with enhanced force-time characteristics, such as rate of force development, mechanical power,

jumping, sprinting, and change of direction ability (25). At all competitive levels, these “high-powered actions” can prove to be the difference between winning and losing; therefore, it is important that players are prescribed appropriate conditioning programs that can improve these components. The challenge in-season is to maintain or improve strength and power qualities with limited time to perform strength and conditioning and with accumulating levels of fatigue. The reader is directed to Strudwick and Walker (24) for a more detailed description as to how this can be achieved.

Increasing strength is also associated with decreased injury rates (18). In European elite teams, muscle injuries constitute 31% of all injuries, with 92% of injuries affecting the 4 major muscle groups of the lower limb: hamstrings (37%), adductors (23%), quadriceps (19%), and calf muscles (13%) (9). In women, most injuries occur to the ankle, knee, and thigh with higher rates of ankle sprains and more serious injuries to the anterior cruciate ligament than in men (17). Several clubs prescribe preventive strategies to reduce the risk of noncontact injury (20), based around FIFA’s 11+ program of core stability, balance, and neuromuscular control exercises. To increase the resilience and robustness of the player, specific exercises to increase the eccentric strength of the hamstrings and adductors have been proposed, with demonstrated changes to muscle architecture (2,13). It has still not been elucidated as to the minimum effective dose to provide preventive benefit; sessions should be strategically planned according to athlete needs in accordance with the on-field program goals.

When a player gets injured, the conditioning coach has a number of roles to play; to train the player to improve his or her ability to tolerate the demands of training and match-play, to make gains in other areas of physical performance, and to strengthen previously injured areas. Although there is no definitive evidence to suggest that

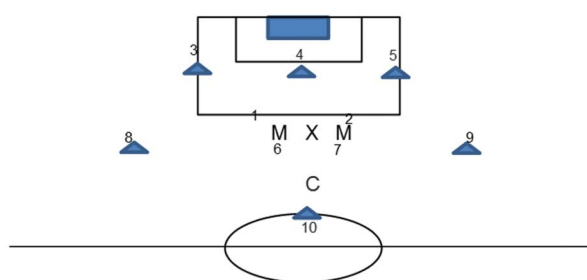


Figure 9. Aerobic high-intensity drill for center back.

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strengthening previously injured areas alone prevents reinjury, it has been demonstrated that increasing eccentric hamstring strength was associated with reduced future hamstring injury risk in a cohort of previously injured Australian soccer players (22). It therefore seems prudent for the strength and conditioning practitioner to develop an athlete sustainability program for players to follow for a period when they return to training and match-play after injury. This period depends on each player's injury history and nature, and for some serious injuries, it may continue indefinitely throughout the player's career.

It is important for the strength and conditioning coach to look for windows within the season such as injury, suspension, or free weeks to maximize off-field conditioning opportunities. Furthermore, as time is extremely limited, maximizing the use of daily warm-ups may provide a key "training"

stimulus for the players. Also, a preparatory tool, the warm-up can be structured to provide benefit for long-term athletic development, providing regular exposures to key fundamental movement patterns to develop movement efficiency. Expanding upon the 11+, an overview of a 20-minute squad "warm-up" circuit conducted before training twice per week is presented in Table 2. Players work for 5 minutes in each group and then move to the next group until all exercises are completed. Exercise selection, sets, repetitions, and resistance will depend upon the age and experience of the player and stage of season.

SQUAD MANAGEMENT

The physical, technical, and psychological demands imposed upon elite players are largely dependent upon team success, with an increased number of games associated with European and domestic Cup Competitions. A

potential weekly training template was presented in Figure 3, but in reality, weekly planning is dictated by the fixture schedule together with the current fitness status of the players, which is often different for each individual. Therefore, it is logical to include flexibility in the conditioning program and tailor weekly templates to the specific requirements at that time, instead of creating a single standard weekly program.

Generally, teams are faced with the following scenarios: 1 match per week or 2 matches per week (frequently this involves playing 3 matches in an 8-day period, i.e., Saturday-Wednesday-Saturday). The latter scenario provides the biggest challenge to the conditioning coach because there requires a delicate balance of training for starters, substitutes, and nonsquad players. There are a large number of strategies that can be used to structure training in this situation; thus, it is not possible to

Table 2
Squad injury reduction warm-up strategy

	Area	Rationale	Sample exercises
Group 1	Glute activation/movement mechanics	Strengthen the glutes, groins, and hamstrings with functional movements using light or no resistance. The inability to activate gluteal group is associated with injuries such as lower back pain, hamstring strain, and anterior knee pain.	Glute mini-band walks, med ball squat, lunge, lateral squat, single leg romanian deadlift.
Group 2	Torso stability/strengthening	To improve the players' ability to transfer force from the ground to the extremities, while preventing uncontrolled movements of spine/pelvis.	Torso stability: front and side plank exercises and variations, posterior stability exercises. Torso strengthening: (a) antiextension exercises; movements of arm and leg with limited movement of torso, (b) rotational and antirotational exercises, producing and resisting against rotary force.
Group 3	Proprioception/landing mechanics	To improve the ability of the body to land and stabilize correctly, yet provide a foundation from which to progress to more explosive plyometric movements. A recurrence of inadequate landings may lead to ankle, knee, and hip joint injury.	Bilateral landings in place, bilateral landings with movement, unilateral landings in place, unilateral landings with movement. Intensity can be increased by increasing height/resistance/surface instability.
Group 4	Hip mobility	The ability to maintain strength through a range of motion is important for soccer players to prevent injury and to provide a platform to produce power.	Hurdle overs (hip flexion/extension), hurdle adduction/abduction (bent/straight leg), resisted leg swings (front-back, side-side), hurdle unders, and slideboard adduction.

Table 3
Training requirements for 3 groups within the team during a congested fixture period

	Playing squad	Substitutes	Non-playing squad
Saturday	Match	Match	High-Intensity aerobic/individual
Sunday	Rest or recovery	Intensive football	Intensive football
Monday	Individual requirements ^a	Speed/power	Speed/power
Tuesday	Reactivity/activation	Reactivity/activation	Reactivity/activation/top-up
Wednesday	Match	Match	Day off/rest
Thursday	Recovery	Speed/power	Speed/power
Friday	Reactivity/activation ^a	Reactivity/activation	Reactivity/activation
Saturday	Match	Match	Individual requirements

^aStrategies adapted depending upon individual status of the player.

detail all potential scenarios. A generic overview of the type of training that each group within the team can perform is detailed in Table 3.

PLAYING SQUAD

When 2 matches are played per week, it is more difficult to perform additional physical training because the majority of the time is dedicated to develop game readiness, tactical preparation, and player freshness. The first decision to be made is whether to give the players a day off after a match or to perform a formal recovery session at the training ground. There is a balancing act between preserving psychological freshness among the squad while also allowing players some quality time with their families and ensuring a competitive focus is maintained. Factors including the number of consecutive days without a day off, previous long-distance travel, timing to the next match, stage of the season, subsequent fixture importance, together with the coach's experience, and knowledge of the wellness status of the squad may be used to formulate a decision.

It is extremely important to regularly monitor players to determine how they respond to the imposed demands of match-play; the 48 hours after a match may be the best time to monitor fatigue status in the team because it is during this time that the greatest

training errors are made. Players respond differently to matches and the accumulation of matches, and thus, some players may be fully recovered yet others may require additional recovery strategies. By accurately accessing the fatigue status of each player (26), flexible and individualized recommendations can be provided to the coaching staff. Once again, it is clear that there is no "one size fits all" approach to be applied in this complex decision-making situation; knowledge of individual player capabilities is crucial.

SUBSTITUTES AND NONPLAYING SQUAD

For the substitutes and nonplaying squad, the day after a game is a crucial conditioning day to ensure that players are physiologically prepared for games. In this situation, training usually consists of high-intensity conditioning work through small-sided games with additional individual drills as appropriate. It is important to ensure that these players receive competitive match minutes in reserve team fixtures, but where that is not possible, individuals must receive "top-up" training in which high-intensity speed and speed-endurance runs are performed at the correct time as highlighted in Figure 5. The reader is directed to Table 2 for an overview of

the high-intensity running and sprint requirements for players by position during a match, to provide the conditioning coach with a guide as to the most important physical parameters to replicate during these sessions. Without this, regular substitutes or nonplaying squad members have the potential to lose their conditioning level over a period, yet these players are important throughout a season from the perspective of squad rotation. Their training load must be sufficient to allow them to cope with the physical requirements of match-play because abrupt and severe increases in workload (due from switching from nonplay to regular play) may enhance the risk of injury occurrence.

Communication between technical and conditioning staff is extremely important with regard to the management of these players because it is vital to maximize conditioning opportunities from a prematch, matchday, and post-match day perspective. If the match-day squad is planned in advance and the conditioning coach is aware, nonsquad players can perform individual conditioning work (on- and/or off-field) after the relatively light prematch day training session. Furthermore, following matches, substitute players can perform high-intensity conditioning work on the pitch, followed by key preventive

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strength and conditioning exercises in the stadium. This is particularly useful if the day following a match is given off as rest, to ensure that no conditioning opportunities are missed.

SUMMARY

As a consequence of the extent of the competitive demands over the course of the season, the traditional planning templates proposed in the literature are of little relevance due to the ensuing fixture congestion. This discussion has centered on combating the difficulties that arise from this scenario that is ever present at the elite level. **It is known that playing 2 matches within a 72-hour period has been associated with reduced high-intensity running performance (21) and significantly increased injury occurrence (8), factors that could result in a team being less successful; strategies therefore are required to combat these issues.** In addition to the key planning strategies focused on, players need to be prescribed appropriate nutritional intake and utilize individualized evidence-based recovery strategies to optimize recovery between matches. Furthermore, it has been suggested that increased physical conditioning level is associated with accelerated recovery (15), and thus, there is a requirement to at least maintain this quality throughout the season.

To deal with fixture congestion and manage an entire season, consideration needs to be given to the rotation of players within the squad. By systematically measuring the external load that players experience in training and match-play, together with monitoring their response to the imposed demands, it is possible to “micromanage” individuals within the squad, providing information to the coach regarding player status. This objective information, together with the match difficulty index (23), can be used to assist in the daily planning of training and in squad selection for upcoming matches. Discussions relating to these objectively driven strategies are beyond the scope of this article; however, they

are clearly important components that support the decision-making processes that blends “art” with “science.”

To summarize, the article has highlighted the processes that are undertaken by conditioning coaches of elite soccer clubs when planning the training program for professional players. Strategies to structure training from a weekly and daily perspective were presented, taking into account the individual needs of players within the constraints of the annual calendar. Ultimately, the aim is to maximize squad availability while optimizing player readiness for matches, ensuring that each player has the physical capability to implement the required tactics successfully.

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